Secure Registration and Login System

Cybersecurity Final Project Documentation

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Hosted Website Link: [Paste PythonAnywhere link here]

1. Hashing Algorithm Used

This project uses the SHA-256 hashing algorithm through Python hashlib.sha256(). During registration, the plain text password is not stored. The system combines password + salt + pepper, then converts the result into a SHA-256 password hash. During login, the hash is recomputed and compared with the stored hash.

2. How Salt Works

A salt is a unique random value generated for every registered user using os.urandom(16). The salt is stored in the database because it is needed during login. Salting helps ensure that two users with the same password still have different password hashes.

3. How Pepper Works

A pepper is a secret value added before hashing. Unlike the salt, the pepper is not stored in the database. This means the database shows only username, password hash, and salt. The pepper adds another protection layer if the database is viewed by an unauthorized person.

4. Password Meter Validation

The password meter validates five requirements: lowercase letter, uppercase letter, number, symbol, and minimum length of 12 characters. The meter displays Weak, Medium, or Strong. Registration accepts only Strong passwords. Example strong password: Cyber@2026Secure.

5. Importance of Strong Passwords

Strong passwords are important because weak passwords can be guessed, brute-forced, or cracked more easily. Requiring length and complexity helps protect user accounts and supports safer authentication practices.

6. Screenshots to Insert

Required Screenshot	Filename
Registration form	01_registration_form.png
Password meter	02_password_meter.png
Successful registration	03_successful_registration.png
Successful login dashboard	04_successful_login_dashboard.png
Failed login attempt	05_failed_login_attempt.png
Database table: username, hashed_password, salt	06_database_table.png
Hosted online system	07_hosted_online_system.png

Note: Replace this starter PDF with a final version containing actual screenshots before submission.

7. Public URL

Hosted website link: [Paste PythonAnywhere link here]

Conclusion

This project demonstrates secure registration and login with password strength validation, hashing, salt, and pepper. It does not store plain text passwords, and the pepper does not appear in the database.